Cytotoxic immune infiltration effects on breast cancer survival and immunotherapy response

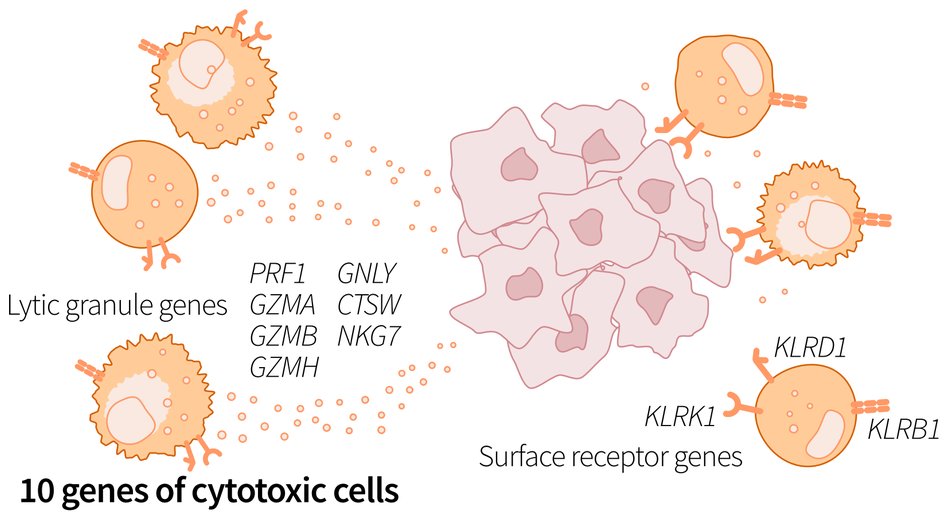
Cross-cohort analysis from METABRIC, TCGA-BRCA & I-SPY2 trial

Chung Yan Yu (Joel) cyy36@cam.ac.uk
MSt in Healthcare Data Science candidate

3 cohorts — METABRIC[1], TCGA-BRCA[2,3], I-SPY2 trial[4]

14 immune cell populations[5]

57 harmonised shared genes[5]



- **Danaher et al. developed a panel of immune population marker genes[4] to score immune infiltration from bulk gene expression.**
- **We screened immune cell populations for associations with survival in two cohorts and with immunotherapy response in a third.**

Identifying representative immune population across two cohorts

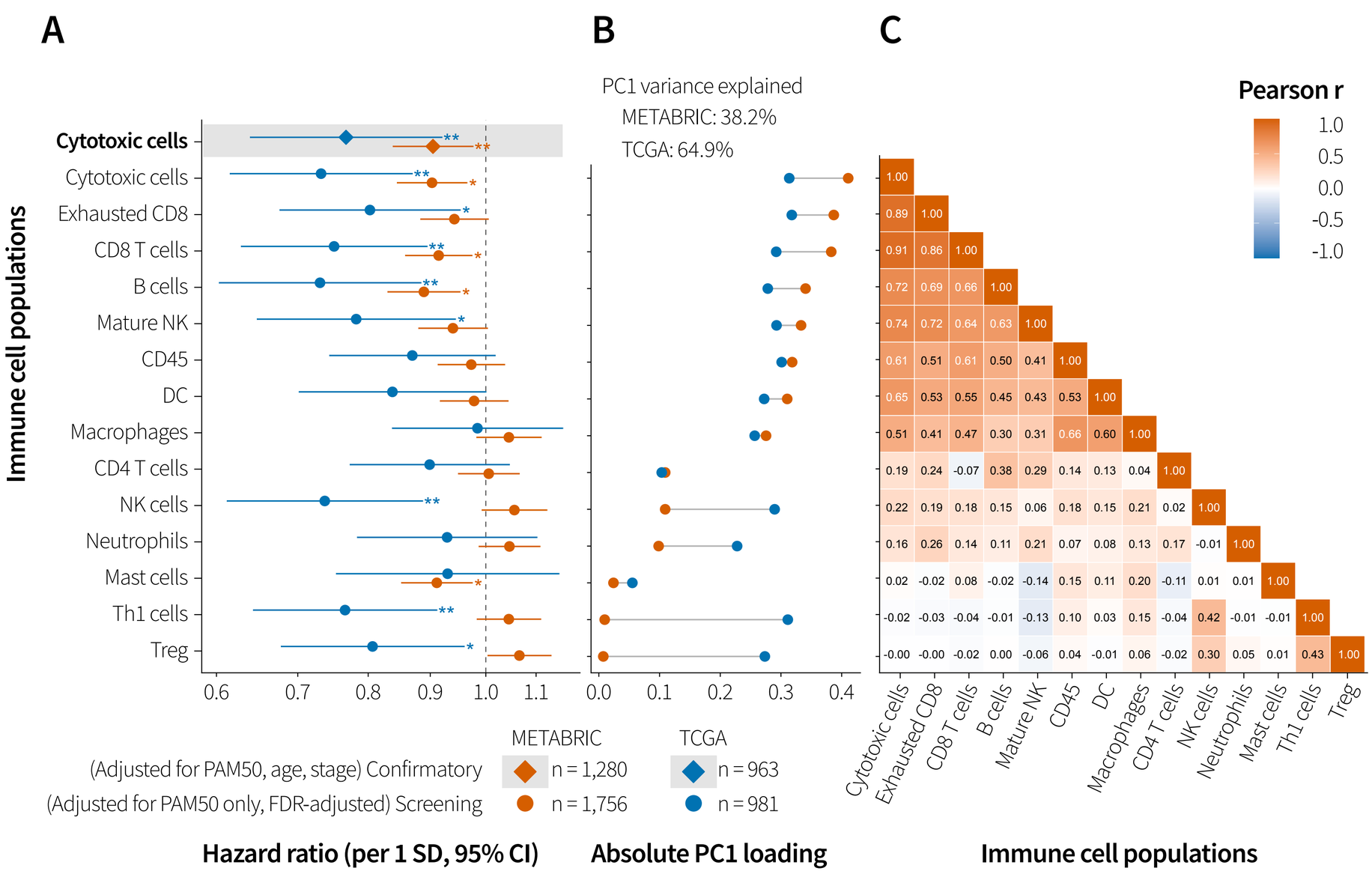


Figure 1 | Identification of cytotoxic cells as a cross-cohort immune axis in breast cancer.
(a) Hazard ratios for 14 immune populations from Cox models in METABRIC and TCGA-BRCA; top row shows fully adjusted cytotoxic result.
(b) Absolute PC1 loadings per immune population across both cohorts.
(c) Pairwise Pearson correlations between population scores in METABRIC, ordered by PC1 loading.

Methods & data sources

METABRIC[1] `cBioPortalData`[6-8]
2,509 clinical
↓ − 529 no expression
1,980
↓ − 244 no OS / PAM50
1,756 screening
↓ − 476 no age / stage
1,280 confirmatory

TCGA-BRCA[2,3] `cBioPortalData`[6-8]
1,084 clinical
↓ − 2 no expression
1,082
↓ − 101 no OS / PAM50
981 screening
↓ − 18 no age / stage
963 confirmatory

I-SPY2[4] - GSE194040 `GEOquery`[9,10]
988 profiled
↓ − 740 other trial arms
248 response
↓ 69 pembrolizumab ↓ 179 control

Cox assumptions
Proportional hazards
- Schoenfeld residual test

Linearity
- Martingale residual test

See bottom QR code for supp info

Immune population scoring
Danaher et al.: 57 genes, 14 populations[5]

Association with survival and response

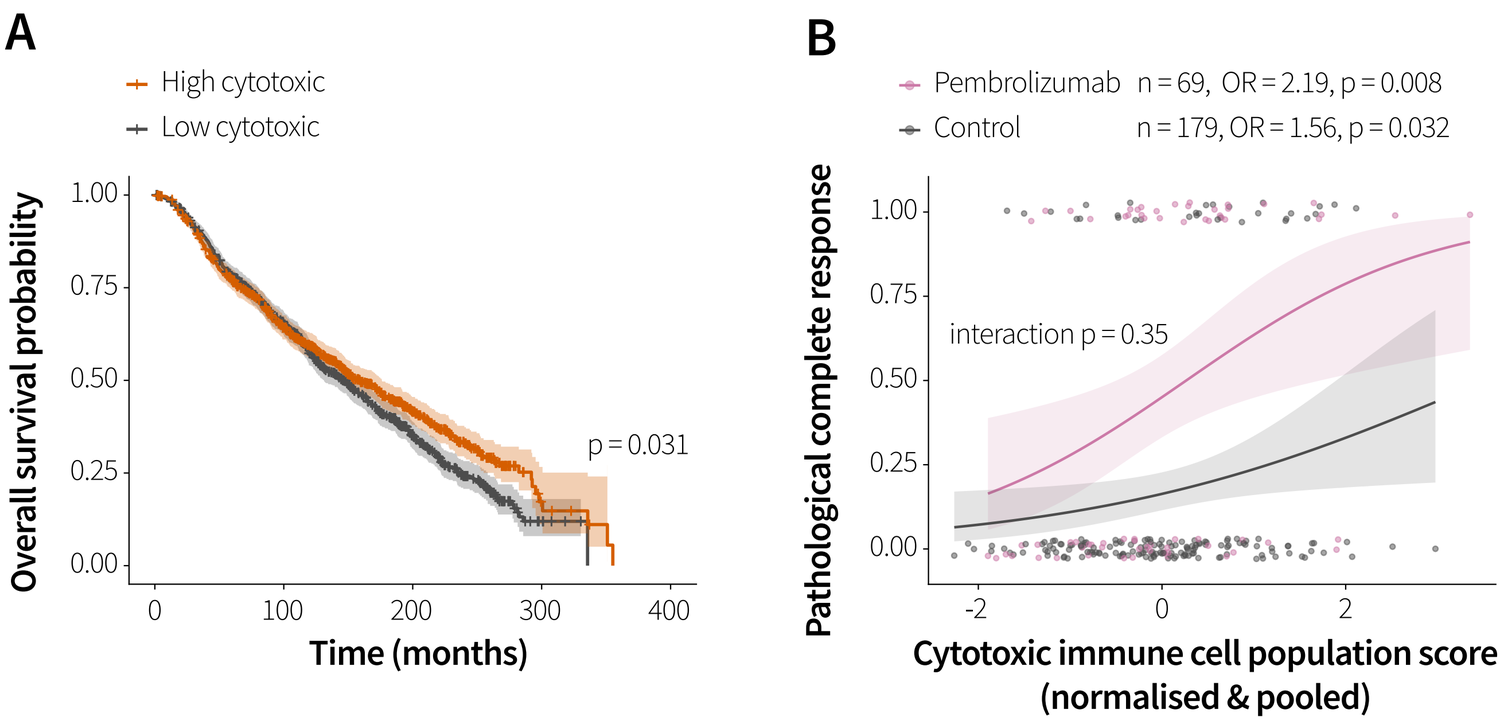
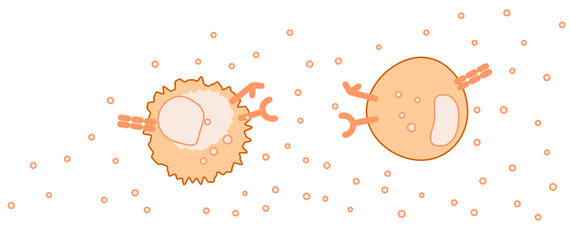


Figure 2 | Cytotoxic infiltration associates with survival and immunotherapy response.
(a) Unadjusted Kaplan–Meier overall survival in METABRIC by median split of cytotoxic score.
(b) Logistic regression of pathological complete response against pooled cytotoxic score in the I-SPY2 pembrolizumab & paclitaxel arm and paclitaxel-only control arm.
All shaded bands are 95% CI.

Cytotoxic infiltration links prognosis to immunotherapy benefit

- **Across METABRIC and TCGA-BRCA, higher cytotoxic scores were independently associated with better overall survival after adjusting for PAM50, age at diagnosis, and tumour stage.**
- **In I-SPY2, higher pre-treatment cytotoxic infiltration associates with pathological complete response in both arms, with a larger point estimate under pembrolizumab than paclitaxel alone.**



References
Supplementary information
Repository
joelyu.github.io/HDS-W4/HDS_04_YuChungYan_2604.html